

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date: 8 March 2026

Team ID: 699ee17ef292113d819d8e62

Project Name: Harnessing Machine Learning for Blood Pressure Analysis

Maximum Marks: 8 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Dataset Collection	USN-1	Collect hypertension datasets including blood pressure readings, age, lifestyle habits and medical history.	3	High	1.Archita Singh , 2.Vanshika Dewan
Sprint-1	Environment Setup	USN-2	Install Python, create virtual environment and install machine learning libraries.	2	High	1.Archita Singh , 2.Vanshika Dewan
Sprint-1	Data Preprocessing	USN-3	Clean dataset, handle missing values and normalize numerical features.	3	High	1.Archita Singh , 2.Vanshika Dewan
Sprint-2	Exploratory Data Analysis	USN-4	Analyze feature distributions and correlations related to hypertension stages.	3	High	1.Archita Singh , 2.Vanshika Dewan
Sprint-2	Feature Selection	USN-5	Select relevant clinical and lifestyle features for prediction.	2	Medium	1.Chandrasekar TM ,2.Bhargav p , 3.Archita Singh , 4.Vanshika Dewan
Sprint-2	Dataset Splitting	USN-6	Split dataset into training and testing sets.	2	High	1.Chandrasekar TM,2.Bhargav p
Sprint-3	Model Implementation	USN-7	Implement machine learning models such as Logistic Regression and Random Forest.	4	High	1.Chandrasekar TM,2.Bhargav p
Sprint-3	Model Training	USN-8	Train models and optimize performance using hyperparameter tuning.	3	High	1.Chandrasekar TM,2.Bhargav p
Sprint-3	Model Evaluation	USN-9	Evaluate models using accuracy, precision, recall and confusion matrix.	2	High	1.Chandrasekar TM,2.Bhargav p
Sprint-4	Prediction Pipeline	USN-10	Build pipeline to predict hypertension stages from patient data.	3	High	1.Chandrasekar TM,2.Bhargav p
Sprint-4	Recommendation System	USN-11	Generate lifestyle recommendations based on prediction results.	2	Medium	1.Chandrasekar TM ,2.Bhargav p , 3.Archita Singh , 4.Vanshika Dewan
Sprint-4	User Interface	USN-12	Develop simple interface for entering patient data and viewing predictions.	3	High	.Chandrasekar TM,2.Bhargav p
Sprint-4	Deployment	USN-13	Deploy the model and application with documentation.	3	High	.Chandrasekar TM,2.Bhargav p

Project Tracker, Velocity & Sprint Schedule

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Feb 2026	01 Mar 2026	20	01 Mar 2026
Sprint-2	20	6 Days	03 Mar 2026	08 Mar 2026	20	08 Mar 2026
Sprint-3	20	6 Days	24 Feb 2026	01 Mar 2026	20	01 Mar 2026
Sprint-4	20	6 Days	03 Mar 2026	08 Mar 2026	20	08 Mar 2026

Velocity

Imagine we have a 10-day sprint duration and the velocity of the team is 20 story points per sprint. Let us calculate the team's average velocity.

$$AV = \frac{velocity}{sprint\ duration}$$

$$AV = \frac{20}{10} = 2$$

Average Velocity = 2 story points per day.

Burndown Chart

A burndown chart is a graphical representation of work left to do versus time. It is commonly used in agile software development methodologies such as Scrum. The chart helps teams track progress and determine whether they are on schedule to complete the work planned for the sprint.

Example representation of Burndown Progress

Day	Remaining Story Points
Day 1	40
Day 2	35
Day 3	30
Day 4	25
Day 5	20
Day 6	15
Day 7	10
Day 8	5
Day 9	0

References

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>
<https://www.atlassian.com/agile/tutorials/burndown-charts>
<https://www.atlassian.com/agile/project-management>
<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>
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